- **Miscellaneous Attacks** [Wei et al., 2023] Collection of various jailbreaking techniques including prompt distractors, using low-resource languages, and instruction-following manipulations.

We filter the generated attacks such that all of the jailbreaks we evaluate our probes on result in a completion that achieves a StrongREJECT (Souly et al. [2024]) score greater than 0.5. We discard all failed jailbreaking attempts.

A.3 Additional results

Fig. 18 shows the transfer between embedding obfuscation attacks designed to jailbreak Llama-3-8b-Instruct between probes. It is the full version of version of Fig. 4c shown in the main paper.

In Fig. 19 we show the performance of probes on the LAT model [Casper et al., 2024] against jailbreaking embedding attacks with and without an obfuscation loss. The trends are very similar to Fig. 6, we can successfully induce obfuscated representations against all probes on the LAT model.

Fig. 20 shows the recall over time during the adversarial training experiment in §3.5 for the ensemble probe. This is the companion figure to Fig. 8a.

B Details on backdoor detection

B.1 Obfuscation training loss functions

Here we provide extra information relevant to section 4.2.

We use the obfuscation loss function

$$\mathcal{L}_{\text{obf}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{x \in \mathcal{D}_{\text{backdoored}}} D_i(x; M) \tag{5}$$

where $D_i(x; M)$ is the anomaly score assigned by detector D_i to the activations $\mathcal{R}(x; M)$ of the sample x from the dataset $\mathcal{D}_{\text{backdoored}}$ of backdoored inputs (and possibly generations too). In other words, the obfuscation loss adjusts the model weights so that backdoored activations are assigned low anomaly scores by the current copies of the detectors $\{D_1, \ldots D_{\text{N}}\}$. Note that this loss function does *not* model the dependence of the detector D_{i} itself on the model, this is taken into account only through iterative updating of the detector rather than through immediate gradients.

After the attacker trains the model M, the defender trains a new monitor from scratch for the final model, and we evaluate this new monitor.

This attack could fail if the model changes too much, so that the monitors D_i become outdated. The new monitor trained by the defender might then perform much better than the attacker's copies. To make this less likely, we have the attacker add an activation retention loss that penalizes deviations from the model's original activations on clean samples:

$$\mathcal{L}_{\text{retain}} = \sum_{x \in \mathcal{D}_{\text{clean}}} \left\| \mathcal{R}(x; M) - \mathcal{R}(x; M_{\text{original}}) \right\|_2 . \tag{6}$$

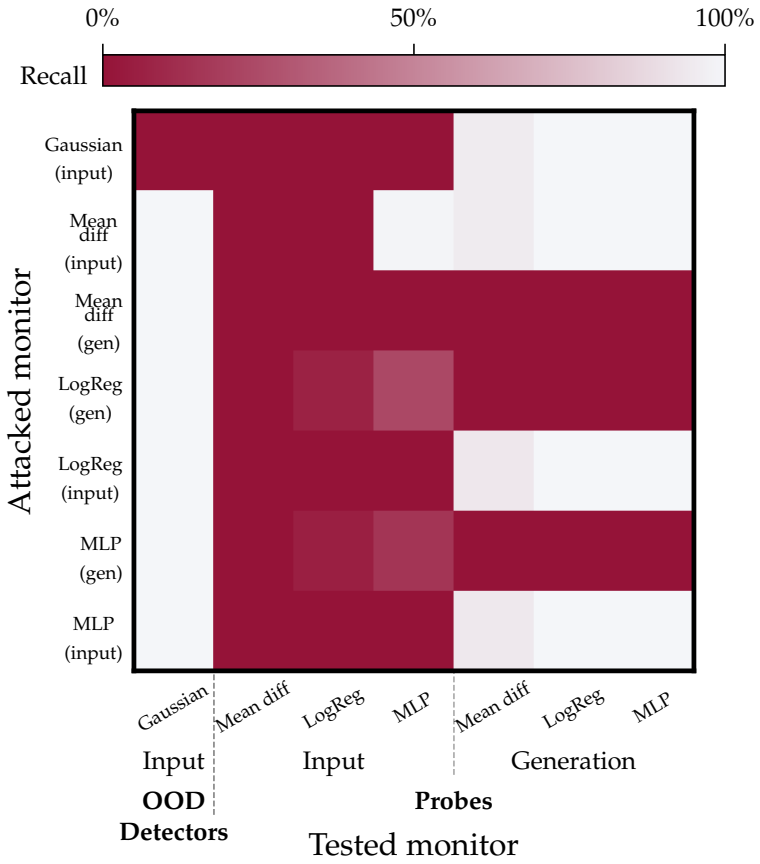

Figure 18: Heatmap showing the performance between obfuscating embedding attacks trained to jailbreak Llama-3-8b-Instruct and fool a given monitors (rows) tested against other monitors (columns). This is the full version of Fig. 4c shown in the main paper.

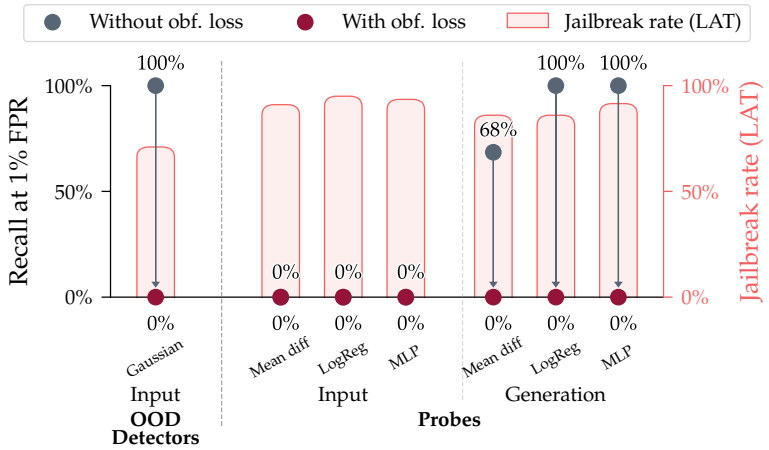

Figure 19: **Obfuscation attacks work against the LAT model**. **Baseline probe recall** against attacks without explicit obfuscation loss are low, often 0% (as shown also in row 3 of Fig. 4b). For probes that perform well on the baseline, we can still obfuscate against them with **obfuscation attacks** while still achieving a high **jailbreaking rate** of over 60%. Note that OOD detector baselines perform better than with the normal model in Fig. 4b, likely because latent adversarial training makes harmful activations more unusual.

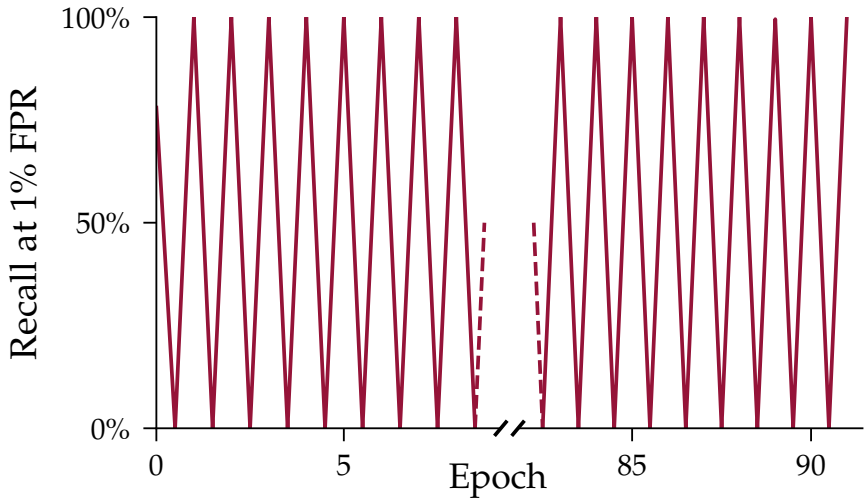

Figure 20: Recall over time for the adversarial training experiment in §3.5 for the Ensemble probe. Companion plot to Fig. 8a.